

Building 101 & 102: AHU S1 Sequence of Operation

Run Conditions - Requested:

The unit shall run whenever:

- Scheduled to run based on time of day schedule or override schedule.
- OR unit control is overridden manually by the operator.

Cold Deck Control:

The cooling supply air temperature setpoint shall be reset based on zone cooling requirements. Zone terminal units shall issue a cooling demand of "1" if the respective zone PID is greater than 90%. Zone terminal units shall issue a cooling demand of "0" if the respective zone PID is less than 90%. If the sum of all zone cooling demands equals zero, the Cold Deck setpoint shall be reset up 1 (Adj.) degree every 1 min (Adj.). If the sum of all zone cooling demands equals one or more, the Cold Deck setpoint shall be reset down 1 (Adj.) degree every 1 min (Adj.).

SA Fan:

The SA fan shall run any time the unit is commanded to run, unless shutdown on safeties. To prevent short cycling, the SA fan shall have a minimum runtime of 5 minutes (adj.).

Alarms shall be provided as follows:

- SA Fan Failure or switched "Off" locally: Commanded on, but the status is off.
- SA Fan in switched "On" locally: Commanded off, but the status is on.

Hot Deck Control:

The heating supply air temperature setpoint shall be reset based on zone heating requirements. Zone terminal units shall issue a heating demand of "1" if the respective zone PID is greater than 90%. Zone terminal units shall issue a heating demand of "0" if the respective zone PID is less than 90%. If the sum of all zone heating demands equals zero, the Hot Deck setpoint shall be reset down 1 (Adj.) degree every 1 min (Adj.). If the sum of all zone heating demands equals one or more, the Hot Deck setpoint shall be reset up 1 (Adj.) degree every 1 min (Adj.).

Freeze Protection:

The unit shall generate an alarm upon receiving a low temperature detection status and start the preheating coil pump and low temperature protection chilled water pump. The OA and EA dampers shall close and the RA damper shall open. The chilled water valve shall be commanded closed and the hot water valve shall be commanded open. The SF will stop.

MA Temperature:

The controller shall monitor the MA temperature and use as required for economizer control.

Monitoring:

Alarms shall be provided as follows:

Zone temperature alarms shall be provided as follows:

- High MA Temp: If the MA temperature is greater than 90°F (adj.).
- Low MA Temp: If the MA temperature is less than 36°F (adj.).
- If the zone temperature is 5°F (adj.) below setpoint.
- If the zone temperature is 5°F (adj.) above setpoint.

Economizer:

The controller shall measure the MA temperature and modulate the economizer dampers in sequence to maintain a setpoint 2°F less than the cold deck temperature setpoint. The OA dampers shall maintain a minimum adjustable position of 15% (adj.) open whenever occupied.

The economizer shall be enabled whenever:

- OA temperature is less than 65°F (adj.).
- AND the SA fan status is on.

END OF SEQUENCE